Characterizing Uncertainty in the Stage–Discharge Rating Curve Using a Bayesian Copula Framework

Niloufar Soheili[1], Shivakumar Balachandran[2], Mahdi Erfani[3], and Mohammad H. Erfani*[3]

[1]Department of Civil Engineering, The City College of New York, New York, NY, USA
[2]Department of Civil Engineering, University of South Carolina, Columbia, SC, USA
[3]Scripps Institution of Oceanography, UC San Diego, La Jolla, CA, USA

Abstract

In river systems, the stage–discharge relationship plays a crucial role in flow characterization and streamflow estimation. Conventionally, this relationship is estimated using rating curves expressed as a fixed functional form. This deterministic approach may not adequately explain or account for the inherent uncertainties in the stage-discharge relationship, which may arise from changes in hydraulic characteristics, morphological alterations in the riverbed, and anthropogenic modifications. This study proposes a stochastic approach that can effectively account for the uncertainties associated with the stage–discharge relationships. For this purpose, a copula-based Bayesian framework is utilized to determine the best-fitting copula family from several candidates to characterize the joint stage–discharge dependence. An analysis of 6,323 USGS gauges across the contiguous United States shows that this stochastic behavior is widespread, not confined to a few sites. Results demonstrate that the copula-based framework can effectively capture the range of stage–discharge variations, which deterministic approaches fail to represent. Applied across sites of varying drainage area, stream order, and hydrologic setting under a single uniform configuration, the framework produced well-calibrated predictive intervals, with 95% coverage between 94.1% and 97.8%. We also demonstrate the practical implications of this approach by integrating the resulting probabilistic rating curves with a conceptual flood inundation model to show how stage uncertainty can impact the inundation extents. These findings highlight that accounting for uncertainty in natural systems yields a more realistic, interpretable representation, particularly for stage–discharge relationships where probabilistic modeling captures the drivers of stochastic behavior.

1 Introduction

Rivers are essential components of hydrologic systems, transporting water through the channels and floodplains. They are naturally dynamic, as they evolve in response to changes in channel morphology, fluctuations in discharge, and sediment transport [1]. These changes alter a river's hydraulic behavior, so the same discharge does not always correspond to the same water level. Understanding the stochastic behavior of rivers can significantly improve flood mitigation, infrastructure design, water-supply planning, and ecological stability [2].

A common simplification to understand flows in rivers involves the use of rating curves to represent the stage–discharge relationship. The rating curve relates stage (water level) to streamflow (discharge) and is typically expressed as a deterministic function, unique to each gauging station for

*Corresponding author: moerfani@ucsd.edu

a given period of time. However, this formulation carries several limitations, as it fails to capture the inherent complexities of sediment transport and hydraulic characteristics. More critically, the stage–discharge relationship is stochastic in nature, and deterministic formulations cannot account for the drivers of this stochasticity. These drivers include, but are not limited to, hysteresis effects during unsteady flow events, particularly those driven by intense precipitation or reservoir releases [3, 4]; complex boundary conditions such as backwater effects [5] and seasonal vegetation growth [6]; and anthropogenic modifications. Consequently, deterministic rating curves fail to represent the inherent uncertainties that are critical for accurate discharge estimation in river systems.

Several studies have improved the representation of stage–discharge relationships. These include recalibrating curve coefficients, adjusting roughness values, and using limited cross-sectional or remotely sensed data to better represent channel geometry [7, 8, 9]. The deterministic approach was further enhanced by incorporating higher-order formulations, such as polynomial, spline-based, or hydraulically informed models, as well as data-driven and machine learning approaches that estimate rating curve parameters from geomorphic and hydrologic attributes at local to continental scales [10, 11, 12, 13, 14]. Despite improving local accuracy in specific settings, these methods result in a unique stage for every discharge, which may not reflect the inherent uncertainty in the stage-discharge relationship. This limitation becomes challenging, particularly at high flows, where direct measurements are scarce, and the curve may have to be extrapolated beyond the observed range [15]. A probabilistic framework that can characterize the uncertainty of the stage-discharge relationship offers more reliable discharge estimations.

To overcome these limitations, likelihood-based and Bayesian frameworks have been developed to explicitly quantify stage–discharge uncertainty by modeling measurement error, hydraulic variability, and complex channel controls [16, 17, 18, 19]. These approaches include joint likelihood formulations that simultaneously estimate rating-curve and flood-frequency parameters [15], Bayesian inference schemes that incorporate hydraulic priors and gauging uncertainty [16], and methods that address backwater effects, unstable channels, and epistemic uncertainty [17, 18, 20, 21]. Collectively, these methods quantify discharge uncertainty and demonstrate that rating-curve uncertainty propagates substantially into downstream hydraulic estimates [15, 19].

Despite these advances, existing probabilistic methods still assume a prescribed functional form for the stage–discharge relationship, typically a single- or multi-segment power law. Uncertainty is then represented around this prescribed function, either through the distribution of its parameters or through additional error terms [16, 21, 17, 18, 22]. Many of these methods require site-specific hydraulic priors and expert knowledge of channel controls to constrain their estimates, which limits their scalability across diverse settings and large river networks [16, 18]. Less attention has been given to frameworks that treat discharge and stage as jointly distributed random variables. Such an approach would characterize their dependence structure directly from observations.

This study introduces a copula-based probabilistic framework for characterizing the stage–discharge relationship in natural river systems. Drawing from multivariate probability theory and Bayesian networks, this approach models discharge and stage as jointly distributed random variables and estimates the conditional distribution of stage given discharge, $f(H|Q)$, directly from field observations. The IFMHA dataset, which provides extensive event-level hydraulic measurements across diverse channel types, serves as the observational basis for fitting and evaluating the proposed framework [23]. To investigate the practical implications of this approach, the resulting probabilistic stage estimates are integrated with the Office of Water Prediction (OWP) Height Above Nearest Drainage (HAND) Flood Inundation Mapping (OWP HAND-FIM) model, demonstrating how uncertainty in the stage–discharge relationship propagates into spatially distributed inundation extents. The framework captures the natural variability of the stage–discharge rela-

tionship directly from observations, and as it requires no prescribed functional form or site-specific hydraulic knowledge, it can be applied across diverse channel types and large river networks.

2 Dataset Description

This study uses the Inventory of Field Measurements of Hydraulic Attributes (IFMHA) as the observational basis for fitting and evaluating the proposed framework [23]. IFMHA is a national dataset of direct streamflow measurements collected by the U.S. Geological Survey (USGS) at streamgages across the contiguous United States (CONUS). Each record pairs a field-measured discharge with the corresponding observed stage, thereby providing direct information about the stage–discharge relationship. The dataset includes repeated measurements at thousands of streamgages spanning diverse hydrologic, climatic, and channel conditions. Its extensive spatial coverage and large number of site-level observations make it well suited for examining whether variability in the stage–discharge relationship is widespread and for developing a probabilistic representation of that variability.

Here, we first evaluated whether the assumption motivating the framework, that comparable discharges can be associated with different observed stages, is supported by the IFMHA measurements. For each streamgage, discharge observations were divided into within-site deciles, and the interquartile range of stage was calculated within each sufficiently populated decile. The site-level metric was then defined as the mean of the retained decile-level stage IQRs. Thus, larger values indicate greater stage variation among measurements belonging to similar, although not identical, discharge ranges. This diagnostic is used only to characterize streamgage behavior across CONUS. The filtering criteria, mathematical formulation, and limitations of the metric are provided in Appendix A.

Figure 1 presents the resulting metric for 6,323 USGS streamgages represented in IFMHA. Values near zero indicate that observations at a streamgage closely follow a single-valued stage–discharge relationship, whereas larger values indicate that substantially different stages were observed within similar ranges of discharge. The numerous streamgages with moderate to large values, shown in orange and red, are distributed across multiple regions of CONUS rather than being confined to a small number of isolated locations. This spatially widespread pattern provides empirical evidence that variability in the stage–discharge relationship is a common feature of the observations and is not merely an artifact of a few anomalous streamgages. The results therefore support the central premise of this study: a deterministic rating relationship cannot fully characterize the observed hydraulic behavior at many sites, motivating a probabilistic framework that represents the conditional distribution of stage for a given discharge.

3 Methodology

This section describes the three main components of our analytical framework. We first present a copula-based approach for jointly modeling the stage–discharge relationship. We then describe the OWP HAND-FIM framework, a conceptual "DEM-based" model to translate discharge or stage estimates into spatially distributed inundation extents. Finally, we provide a brief overview of Gaussian Process Regression (GPR), which is used as a baseline model to compare the performance of the proposed copula-based framework.

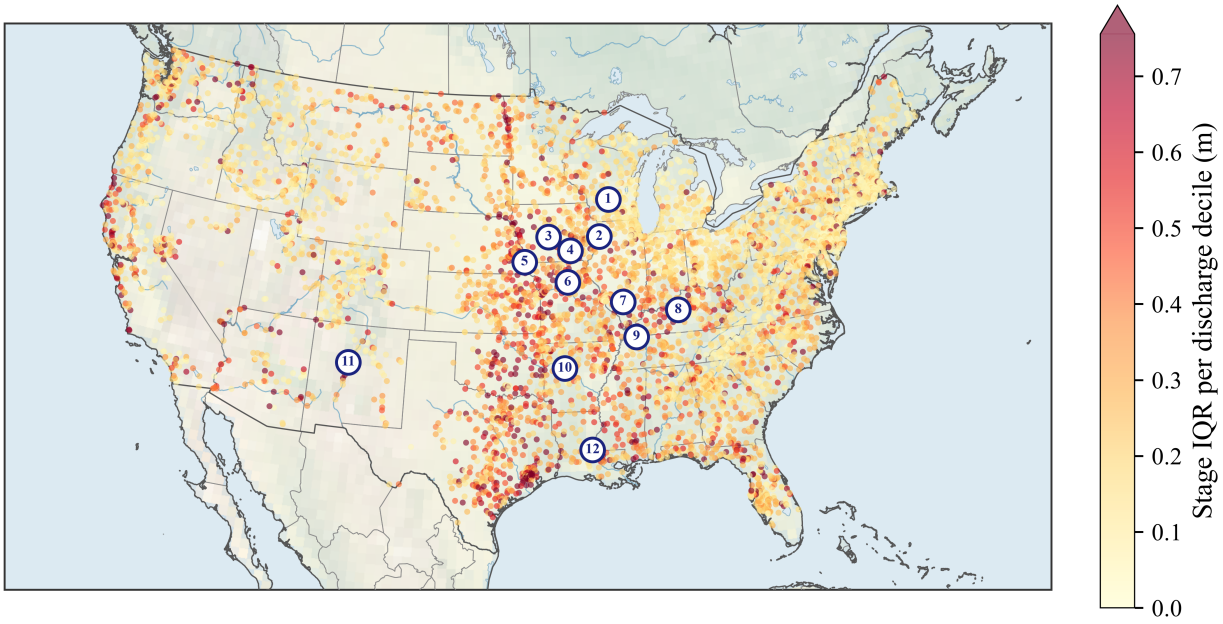


Figure 1: Site-level stage–discharge variability across 6,323 USGS gauges in the IFMHA dataset.

3.1 Stochastic Characterization of Hydraulic Relationships

The Q–H relationship is stochastic in nature, and the modeling framework therefore utilizes a probabilistic representation to remain physically consistent [24, 25]. This relationship is characterized through the conditional distribution $f(H \mid Q)$, derived from the joint bivariate distribution $f(Q, H)$, which captures the nonlinear dependence between discharge and stage directly from observations.

$$f(H \mid Q) = \frac{f(Q, H)}{f(Q)} \tag{1}$$

3.2 Copula-Based Joint Modeling

Copula theory offers a flexible framework for constructing the joint distribution $f(Q, H)$ by separating marginal behavior from dependence structure [26]. According to Sklar's Theorem [27], the joint CDF of discharge and stage can be written as:

$$F(q, h) = C(F_Q(q),\, F_H(h)) \tag{2}$$

where F_Q and F_H are the marginal CDFs of Q and H, and C is a copula that captures all dependence between the two variables. The probability integral transform (PIT), defined as $u = F_Q(q)$ and $v = F_H(h)$, maps each variable to the unit interval $[0, 1]$ and isolates the dependence structure from the marginal scales of Q and H. The bivariate copula density is then obtained as [28]:

$$c(u, v) = \frac{\partial^2 C(u, v)}{\partial u\, \partial v} \tag{3}$$

The joint density function can accordingly be decomposed as [29]:

$$f(q,h) = c(u,v)\, f_Q(q)\, f_H(h) \tag{4}$$

where f_Q and f_H are the marginal probability density functions of Q and H, corresponding to the CDFs F_Q and F_H defined above. By substituting this expression into the definition of conditional probability, the conditional density of stage given a specific discharge level is derived [29, 30]:

$$f(h \mid q) = \frac{c(u,v)\, f_Q(q)\, f_H(h)}{f_Q(q)} = c(u,v)\, f_H(h) \tag{5}$$

The conditional distribution $f(h \mid q)$ is the key quantity in this framework, providing the full distribution of stage at any given discharge and forming the basis of the probabilistic rating curve. The other components of the framework, including marginal distribution selection, copula selection, and performance evaluation metrics, are described in detail in Appendix B. It is worth noting that copula-based modeling requires input observations to be independent and identically distributed (i.i.d.) samples from a stationary joint distribution. The (Q, H) pairs used here are field measurements with temporal spacing typically long relative to the hydraulic response time of the channel. They can therefore reasonably be treated as independent. To further reduce short-term autocorrelation, measurements taken on the same calendar day were averaged into a single value, and a minimum three-day gap was maintained between successive observations. This ensures that each (Q, H) pair represents a distinct hydraulic event.

3.3 Flood Inundation Mapping

We demonstrate the practical implications of the proposed probabilistic framework by integrating the resulting rating curves with the OWP HAND-FIM framework. The National Oceanic and Atmospheric Administration (NOAA) Office of Water Prediction (OWP) Height Above Nearest Drainage Flood Inundation Mapping model (OWP HAND-FIM) is a rating curve-based flood mapping tool. It provides a well-established platform to translate and utilize probabilistic stage estiamtes to a potential real world application of flood mapping. The HAND approach relates water-surface elevation to inundation extent by using the relative elevation of the terrain and the nearest drainage line [31].

In the operational OWP HAND–FIM workflow, stage values are typically obtained from a deterministic Synthetic Rating Curve (SRC). We generalize this step by replacing the SRC stage with percentile estimates derived from $f(H \mid Q)$. Mapping each percentile through the OWP HAND–FIM framework yields a corresponding inundation extent. This approach can effectively capture the uncertainty in the stage–discharge relationship. A major advantage of this method is its flexibility: Both the number of maps and the specific percentiles chosen can be efficiently modeled to generate mutiple flood inundation maps for different levels of risk.

3.4 Gaussian Process Regression

Gaussian Process Regression (GPR) is a non-parametric, Bayesian approach to regression that can flexibly model complex relationships between input and output variables [32]. In this study, GPR is used as a baseline model to compare the performance of the proposed copula-based framework. GPR assumes that the observed data can be modeled as a realization of a Gaussian process, which is defined by a mean function and a covariance function (kernel). The choice of kernel determines the smoothness and structure of the function being modeled. In this study, we use a radial basis function (RBF) kernel that includes a constant white-noise term. Because this noise term is constant across

all discharges, the noise is homoscedastic, and the resulting predictive distribution is Gaussian and symmetric at every discharge. The GPR model is trained on the same (Q, H) pairs used for the copula-based framework, and the predictive distribution of stage given discharge is obtained. The performance of GPR is then evaluated using the same metrics as the copula-based framework, allowing for a direct comparison of the two approaches in terms of their ability to capture the uncertainty in the stage–discharge relationship.

4 Results

4.1 Spatial Variability in Rating Curve Behavior

According to the analysis presented in Section 2, the stage–discharge relationship exhibits substantial variability across many USGS streamgages in the IFMHA dataset. This variability is examined in relation to the channel and physiographic characteristics based on the fluvial hydraulic and geometry attributes presented in the IFMHA dataset (Table 1). To further illustrate this variability, we selected twelve representative sites with different range of variability (numbered in Figure 1). Their observed rating curves are shown in Figure 2.

Table 1: Factors associated with stage–discharge variability.

Category	Factor	Relationship to variability
Geomorphic control	Channel material[a]	Fine, mobile substrate (silt, sand) → more variable
	Channel bed stability	Soft, mobile bed → more variable
Hydraulic control	Channel slope	Lower gradient → more variable
Network-scale control	Drainage area	Larger drainage area → more variable
	Stream order	Higher stream order → more variable
Data quality	Measurement method	More reliance on indirect methods (cableway, bridge, boat) → more variable

[a]Silt, sand, gravel, cobble, boulder.

Channel material shows the clearest link: gauges on silt or sand substrate have more than double the median variability of gauges on boulder- or bedrock-controlled channels ($0.26 - 0.31\,m$ versus $0.12 - 0.14\,m$). Beds of fine, mobile material change shape between measurement visits, as high flows scour sediment away and later flows deposit it again. As a result, the same discharge can produce a different stage on different dates, simply because the bed shape underneath it is no longer the same. Most of the twelve sites examined in this study sit on such beds.

Channel slope shows a similar pattern. Low-gradient channels amplify the sensitivity of stage to discharge. The energy slope enters the governing open-channel flow equations as a square-root term in the denominator. A small change from backwater, an unsteady flood wave, or minor roughness variation can therefore produce a disproportionately large stage response at a given discharge. Larger, higher-order rivers show the same tendency. They are more prone to backwater influence and to hysteresis between the rising and falling limbs of a flood wave. Site 9 drains the largest and highest-order basin among the twelve sites studied here and shows correspondingly high variability; Among the twelve, site 11, with a comparatively steeper reach, shows lower variability.

These factors are not independent. Low-gradient, high-order rivers with large drainage areas tend to also have fine-grained, mobile channel beds. This is why the geographic pattern in Figure 1

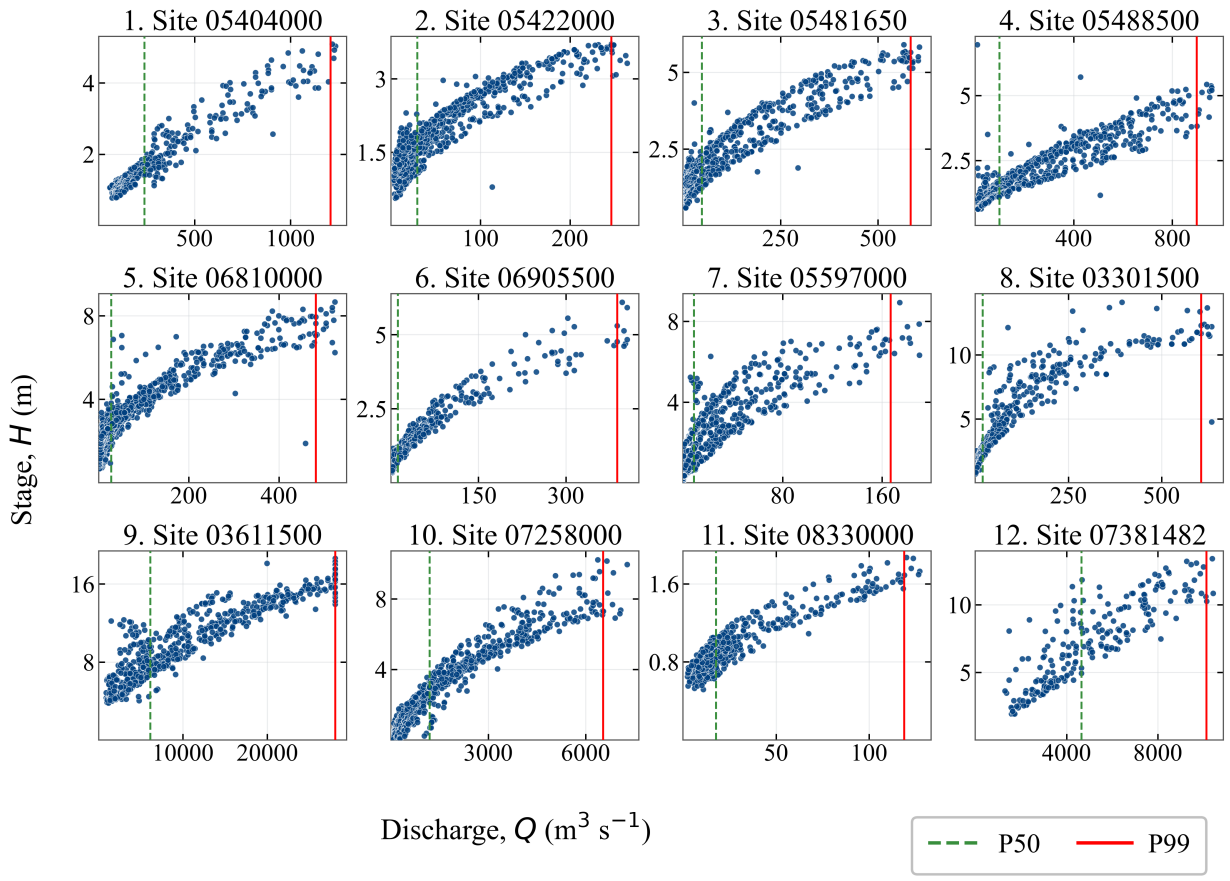


Figure 2: Observed stage–discharge rating curves for the 12 sites numbered in Figure 1.

aligns closely with the alluvial lowland regions of the Great Plains and the Mississippi basin. The steep, bedrock- and boulder-controlled channels of the Appalachian and Rocky Mountain provinces show the opposite pattern.

A secondary, non-physical factor likely also contributes. At high flow, measurement crews at these same gauges shift disproportionately from wading to indirect methods such as cableway, bridge, or moving-boat measurement. Site 7 is a clear example where crews rely almost entirely on bridge and cableway measurement once flow rises above its typical range. These methods carry greater uncertainty than direct wading measurement. Part of the high variability at high-discharge, alluvial-river gauges therefore likely reflects the genuine physical processes described above, and part of it likely reflects this added measurement uncertainty.

4.2 Stochastic Rating Curve Representation

Figure 3 presents the results of the proposed framework applied to the sites discussed previously. Results are shown as both 50% and 90% predictive intervals. Under the more conservative approach (90% interval), the bands encompass the high-flow observations at nearly all sites. In a few instances such as Site 07258000, where some extreme values fall outside the 90% band, coverage can be improved by using a higher-coverage predictive interval (e.g., 95%) or selecting a more conservative copula family. We intentionally retain a single, uniform configuration across all sites to highlight the transferability and generalizability of the proposed framework. The gray points, visible for

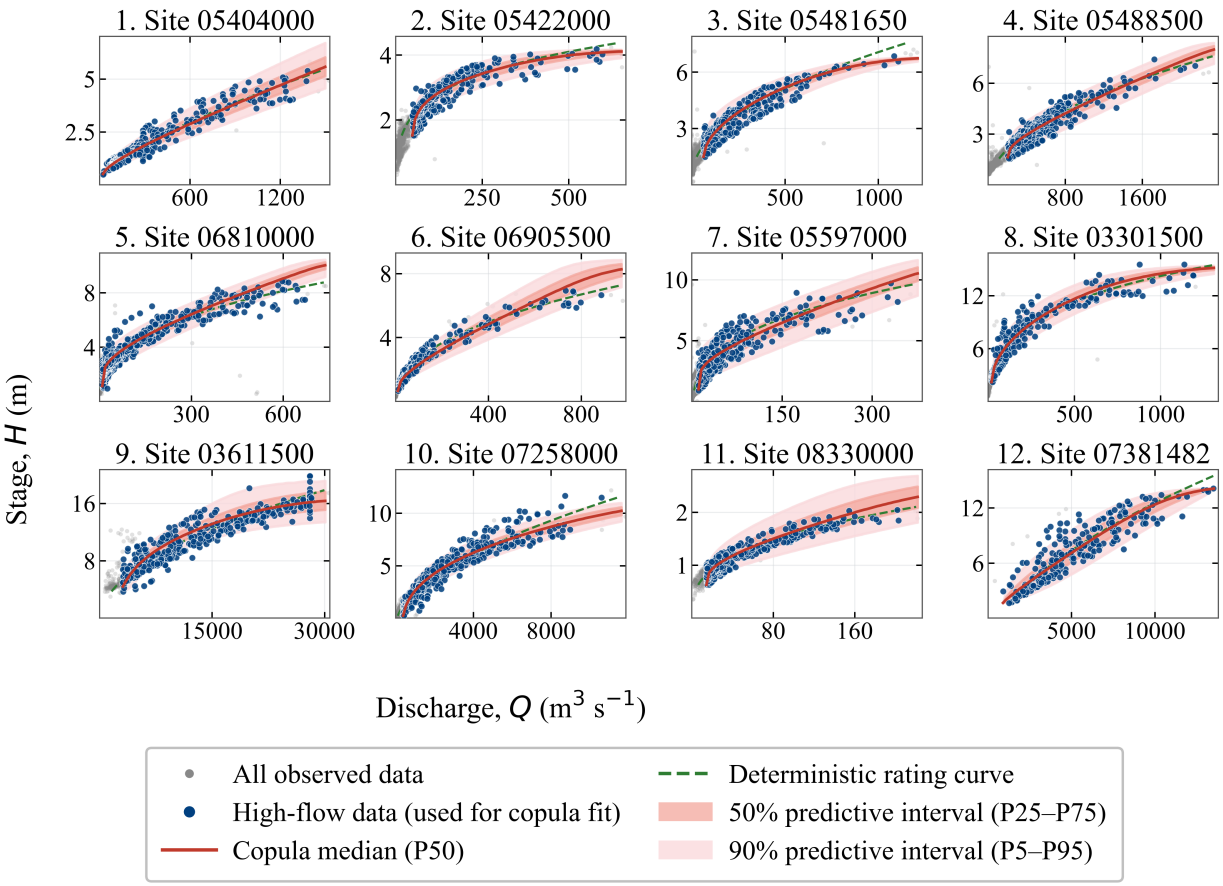


Figure 3: Probabilistic rating curves from the copula-based framework applied to representative IFMHA stations.

example at Site 05481650 and Site 08330000, are the low- and mid-flow observations. They are shown for context but are excluded from the copula fit, which uses only the high-flow subset shown in dark blue. Low flow is excluded because it is not the regime this framework targets: it is well constrained, measured directly, and shows little variability. High flow, by contrast, is where predictive uncertainty is largest. Including the low-flow points would also weaken the fit itself, since they dominate the record in number and would pull the marginal distributions and rank correlation away from the high-flow tail. The rating curve (green dashed line) is still fit to the full record.

4.3 Case Studies

The proposed framework is examined in detail at three USGS streamgages: the Little Sioux River at Correctionville (USGS 06606600), the Des Moines River near Saylorville (USGS 05481650), and the Brazos River at Waco (USGS 08096500). First, the predictive performance of the proposed copula-based framework is benchmarked against Gaussian Process Regression (GPR). Next, its practical implications for flood mapping are demonstrated by generating inundation maps for stage values corresponding to different conditional percentiles at a given discharge. This analysis shows how uncertainty in probabilistic stage estimates propagates into differences in predicted flood extent. Additional site characteristics, marginal-distribution fitting, and copula-family selection are provided in Appendix C.

At the two Iowa sites (Figure 4a, b), the median follows the deterministic rating curve closely across the full discharge range. At the Brazos River (Figure 4c), the two diverge at high flow: the deterministic curve continues to rise, whereas the copula median follows the trend in the observed high-flow data. This divergence reflects the truncated Gaussian marginal fitted to stage at the Brazos River (Appendix C). Its finite upper bound constrains the copula-based stage predictions, whereas the power-law rating curve remains unbounded.

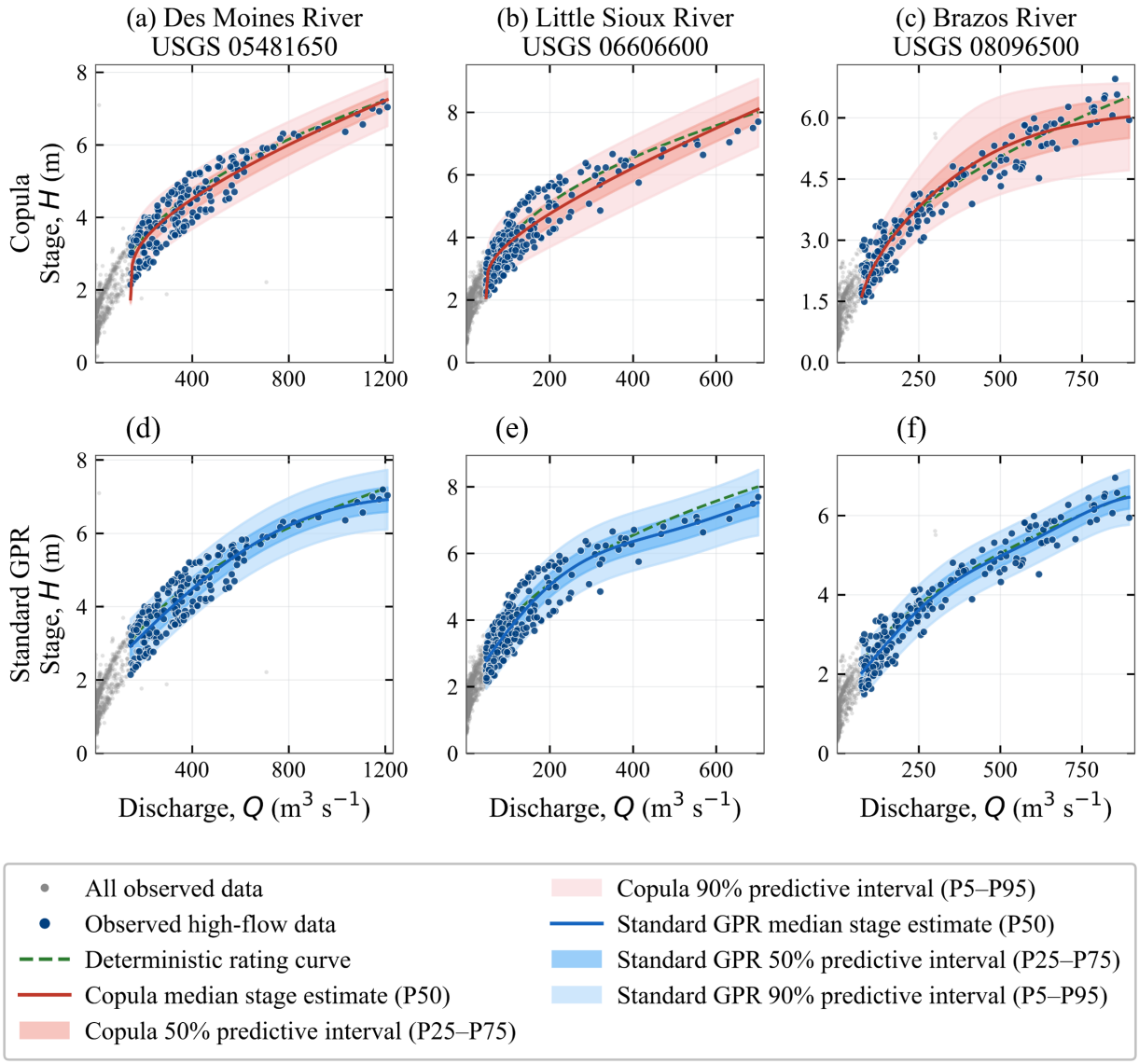


Figure 4: Probabilistic rating curves for the three case-study sites. First row: copula-based probabilistic rating curves; second row: GPR-based probabilistic rating curves.

The proposed framework is compared against GPR at three sites (05481650, 06606600, 08096500), using a standard formulation with homoscedastic noise. Both are fit to the same high-flow subset and the same configuration used for the copula. Both models are scored on 95% interval coverage, average interval width, and the Continuous Ranked Probability Score (CRPS), which increases when predictions are miscalibrated or intervals are wider than necessary (see Table 2).

Table 2: Predictive performance of the copula framework versus standard GPR at each site.

USGS Site	Copula			Standard GPR		
Number	Cov. (%)	Width (m)	CRPS (m)	Cov. (%)	Width (m)	CRPS (m)
05481650	99.1	1.843	0.274	100.0	1.790	0.264
06606600	95.5	2.204	0.326	99.1	2.046	0.297
08096500	94.5	1.945	0.245	96.5	1.496	0.213

Both methods achieve coverage close to the nominal 95% level, indicating that the copula's predictive intervals are well calibrated. The standard GPR attains the lowest CRPS and narrowest intervals at every site, with the copula performing comparably but not better. A Gaussian predictive distribution is symmetric by construction. The GPR's upper and lower half-intervals are therefore equal, and its conditional skewness is zero at every discharge, regardless of kernel or tuning. This symmetry becomes more visible when comparing the copula-based probabilistic rating curve (Figure 4c) with its corresponding GPR-based rating curve (Figure 4f), where the near-uniform bands at high discharge values highlight this difference. The copula is not subject to this constraint. At all three sites, its conditional distribution is right-skewed at low discharge (skewness up to $+1.1$) and left-skewed at high discharge (as low as -0.7). This is consistent with stage being bounded near the channel bed at low flow, and with the rating curve flattening at high flow. No GPR configuration reproduces this reversal.

The copula offers a second advantage that follows from how it is built. It models the joint distribution of discharge and stage. Both conditionals, $f(H \mid Q)$ and $f(Q \mid H)$, therefore follow from a single fit. The two serve opposite operational needs. At a gauging station, stage is monitored continuously and discharge is inferred from it. This is the task a conventional rating curve performs, and it calls for $f(Q \mid H)$. In flood inundation modeling the direction reverses. Discharge comes from a hydrologic model or forecast, and the stage it would produce must be estimated. This calls for $f(H \mid Q)$. A regression, however, is fit in one direction only. Obtaining the reverse conditional would require a second, independently fit model, with no guarantee of consistency between the two. The copula supplies both from the same fit, and the two are consistent by construction. The copula framework matches a well-tuned GPR baseline on calibration and sharpness. It additionally offers an asymmetric, physically consistent predictive distribution and native bidirectional prediction, neither available from a regression-based baseline.

4.3.1 Flood Inundation Mapping with Stochastic Rating Curve

For each of the three case-study sites, the probabilistic rating curve is paired with the inundation extents at the 5th and 95th percentile stages as well as its corresponding deterministic stage obtained from synthetic rating curve (SRC). It's worth noting that the 90% prediction intervals used to produce the flood inundation maps come from a fitted copula distribution that has a appropriate spread metric and are not simply overestimating the range of uncertainty.

At the Little Sioux River (Figure 5), a discharge of $Q = 400$ m^3 s^{-1} corresponds to a stage range of $H_{P05} = 5.02$ m to $H_{P95} = 7.23$ m (i.e., a spread of 2.21 m). The lower bound stays confined to the channel corridor while the upper bound extends across the floodplain and covers the town of Correctionville. The OWP HAND-FIM SRC ($H = 5.66$ m) falls between H_{P05} and H_{P25}, near the 21st percentile.

At the Des Moines River (Figure 6), a discharge of $Q = 600$ m^3 s^{-1} yields $H_{P05} = 4.47$ m and $H_{P95} = 5.99$ m (i.e., a spread of 1.52 m). While small, this 1.52 m difference produces a

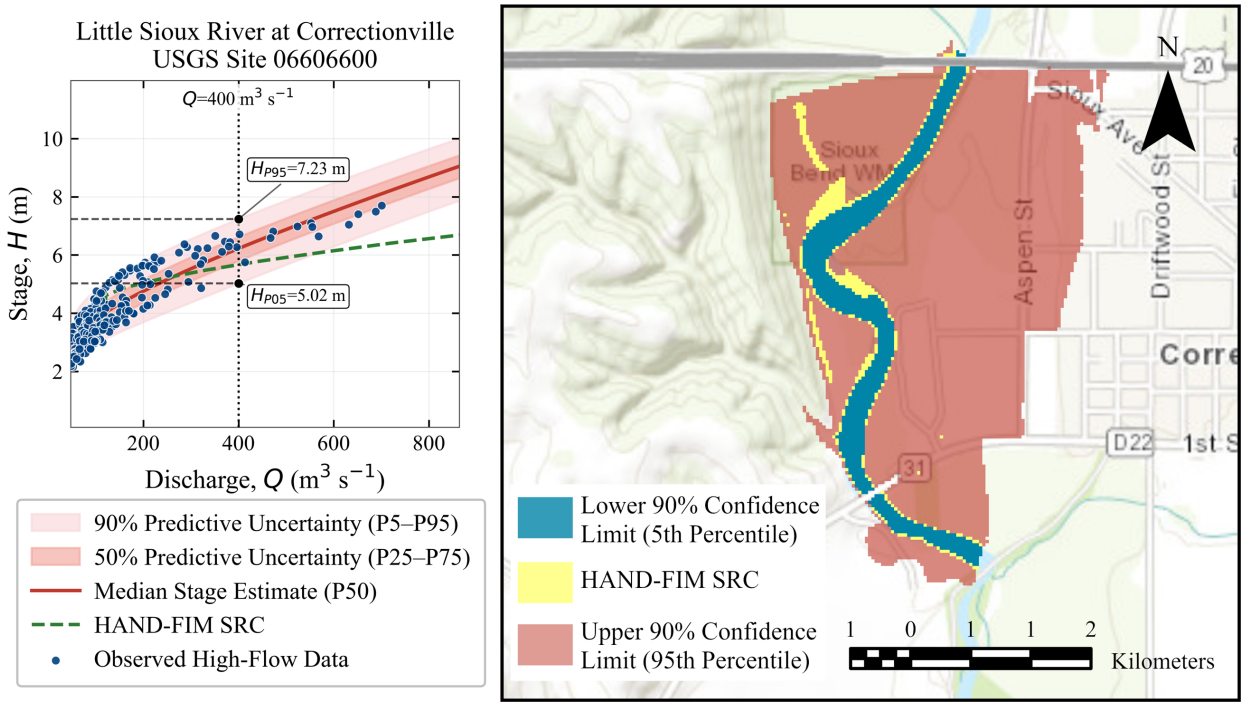


Figure 5: Little Sioux River at Correctionville (USGS 06606600) at $Q = 400$ m^3 s^{-1}. Left: probabilistic rating curve with the 5th/95th percentile stage estimates. Right: corresponding FIMs.

large change in inundated area: the flat Iowa terrain offers little topographic resistance, so the flood extent expands substantially between the lower and upper bounds. The SRC ($H = 5.17$ m) falls between H_{P25} and H_{P50}, near the 38th percentile, closer to the median than at the Little Sioux River. Away from the main channel, both the SRC and upper-bound extents include isolated flooded patches over agricultural fields with no visible connection to the river. This is a known limitation of HAND: it assigns inundation by elevation threshold alone, without checking hydraulic connectivity.

At the Brazos River (Figure 7), a discharge of $Q = 600$ m^3 s^{-1} produces the widest stage range among the three sites. The 90% predictive interval extends from $H_{P05} = 4.36$ m to $H_{P95} = 6.70$ m, corresponding to a spread of 2.34 m. This stage uncertainty does not produce a large difference in total simulated flood extent between the lower- and upper-bound scenarios; however, much of the additional area inundated under the upper-bound scenario falls within developed parts of Waco. This indicates that, even when the areal difference is modest, uncertainty in stage estimates can materially affect estimates of urban flood exposure. The SRC ($H = 5.29$ m) falls between H_{P25} and H_{P50}, near the 37th percentile.

Across all three sites, the SRC estimate fell below the copula median, corresponding to approximately the 21st to 38th conditional percentiles. Thus, for these case studies, the SRC cannot be characterized as conservative from a flood-hazard perspective. This finding does not imply that the SRC is unreliable; rather, a deterministic hydraulic estimate represents a single point within a range of plausible stage outcomes and does not, by itself, reveal its position within that range. The Brazos River case further demonstrates that stage uncertainty does not necessarily translate proportionally into inundation uncertainty: despite its wide stage range, the corresponding change in flood extent was relatively small because the local floodplain topography constrained lateral expansion. Taken together, these findings suggest that site-specific conditions influence both the position

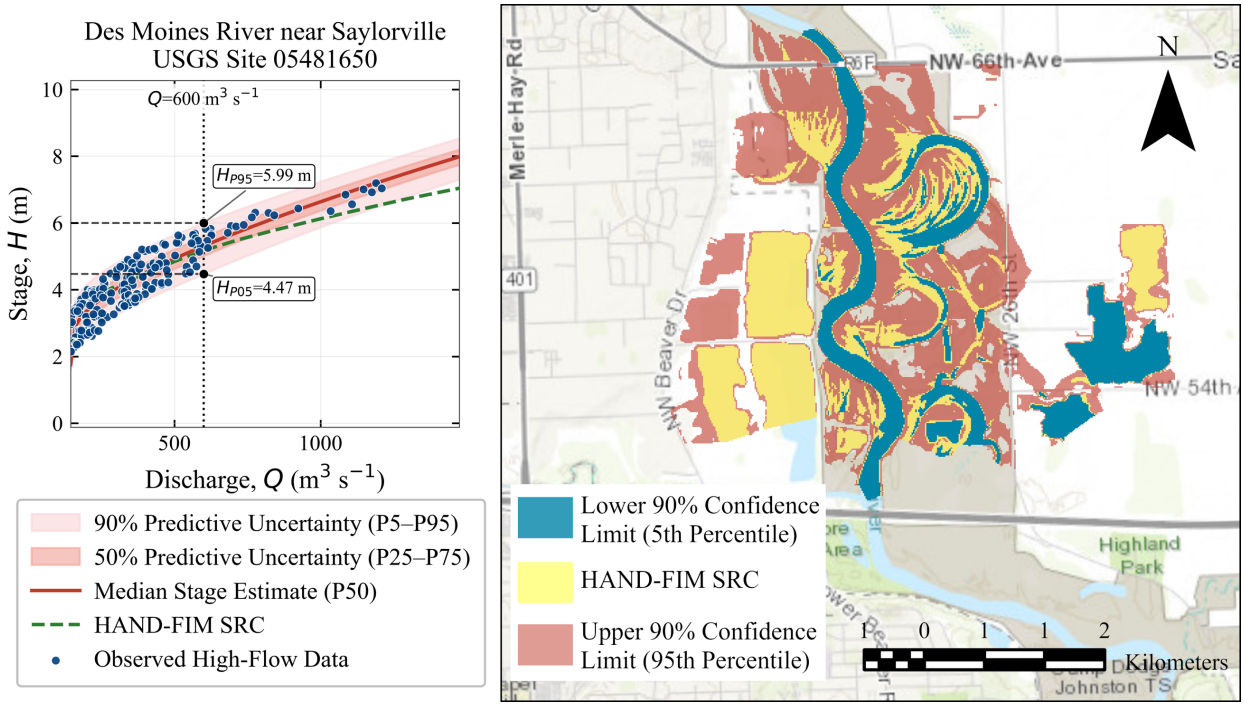


Figure 6: Des Moines River near Saylorville (USGS 05481650) at $Q = 600$ m^3 s^{-1}. Left: probabilistic rating curve with the 5th/95th percentile stage estimates. Right: corresponding FIMs.

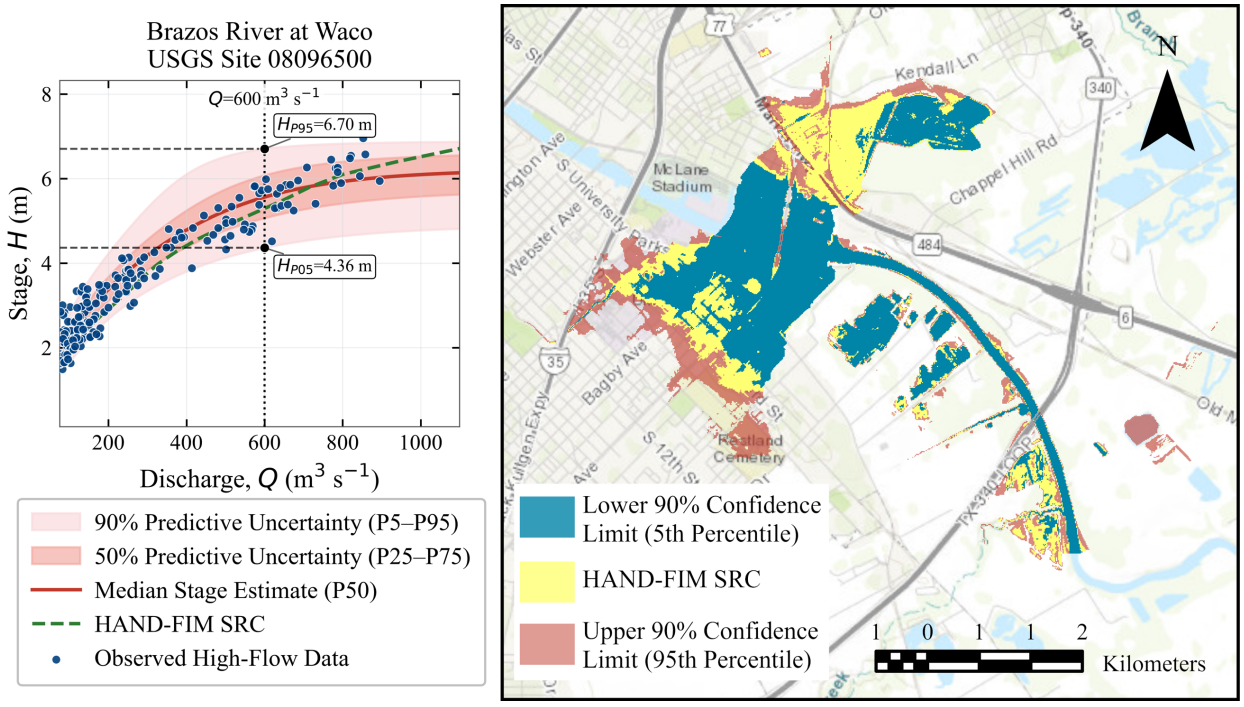


Figure 7: Brazos River at Waco (USGS 08096500) at $Q = 600$ m^3 s^{-1}. Left: probabilistic rating curve with the 5th/95th percentile stage estimates. Right: corresponding FIMs.

of the SRC within the probabilistic stage distribution and the extent to which stage uncertainty affects mapped inundation.

5 Conclusion

Accurately characterizing the stage–discharge relationship, and the uncertainties inherent to it, has long been a challenge in river hydraulics. This relationship is conventionally represented by a deterministic rating curve, yet it is inherently stochastic and cannot be fully captured by a single deterministic function. Here we introduced a copula-based probabilistic framework, drawing from multivariate probability theory, to model this relationship directly from field observations. The approach treats discharge and stage as jointly distributed random variables and estimates the conditional distribution of stage given discharge, $f(H \mid Q)$, without assuming a functional form or requiring site-specific hydraulic knowledge.

A CONUS-scale analysis of 6,323 USGS gauges showed that this stochasticity is systematic rather than site-specific. Stage variability at fixed discharge is highest in the low-gradient alluvial region, and lowest in the steep, bedrock-controlled channels. This is the variability the framework is designed to represent. The framework was applied across sites spanning a range of drainage area, stream order, and hydrologic setting. It produced well-calibrated predictive intervals, with 95% coverage between 94.1% and 97.8%. These results used a single uniform configuration rather than site-by-site calibration, which supports its transferability across such diverse settings. The framework's implications extend beyond the rating curve itself. Propagating the probabilistic stage estimates through the OWP HAND–FIM model showed that a single discharge yields a range of plausible inundation extents. This propagation is non-proportional and depends on local floodplain geometry.

Although the proposed framework requires no prior information about the hydraulic characteristics of a site, which is an advantage, it fits statistical models for both the marginals and the dependence structure. It therefore carries more degrees of freedom than a deterministic power law and needs a larger number of reliable observations to be well constrained. Future work could broaden the framework's reach. Regionalizing copula parameters using catchment and channel attributes such as drainage area, slope, and stream order would extend the approach to ungauged sites. The probabilistic estimates could also be integrated into NWM-based inundation mapping as uncertainty-aware complements to synthetic rating curves, or used to generate synthetic stage–discharge pairs where observations are sparse. The resulting distributions could further serve as an observation-error model for calibrating rainfall-runoff models directly in stage space, reducing the parameter bias introduced by ignoring rating-curve uncertainty [33]. More broadly, acknowledging the stochastic nature of the stage-discharge relationship supports more realistic and risk-aware decisions in streamflow forecasting, infrastructure design, sediment and water-quality modeling, and water-resources planning.

References

[1] Bruce L. Rhoads. *River Dynamics: Geomorphology to Support Management*. Cambridge University Press, 2020.

[2] Nigel W. Arnell and Simon N. Gosling. The impacts of climate change on river flood risk at the global scale. *Climatic Change*, 134(3):387–401, 2016.

[3] E. Perret, M. Lang, and J. Le Coz. A framework for detecting stage–discharge hysteresis due to flow unsteadiness: Application to France's national hydrometry network. *Journal of Hydrology*, 608:127567, 2022.

[4] D. L. Fread. Computation of stage-discharge relationships affected by unsteady flow. *JAWRA Journal of the American Water Resources Association*, 11(2):213–228, 1975.

[5] H. Hidayat, B. Vermeulen, M. G. Sassi, P. J. J. F. Torfs, and A. J. F. Hoitink. Discharge estimation in a backwater affected meandering river. *Hydrology and Earth System Sciences*, 15(8):2717–2728, 2011.

[6] E. Perret, B. Renard, and J. Le Coz. A rating curve model accounting for cyclic stage-discharge shifts due to seasonal aquatic vegetation. *Water Resources Research*, 57(3):e2020WR027745, 2021.

[7] Vijay P. Singh. Derivation of rating curves using entropy theory. *Transactions of the ASABE*, 53(6):1811–1821, 2010.

[8] S. Barbetta, M. Franchini, F. Melone, and T. Moramarco. Enhancement and comprehensive evaluation of the rating curve model for different river sites. *Journal of Hydrology*, 464–465:376–387, 2012.

[9] Feng Pan, Cheng Wang, and X. Xi. Constructing river stage–discharge rating curves using remotely sensed river cross-sectional inundation areas and river bathymetry. *Journal of Hydrology*, 540:670–687, 2016.

[10] J. D. Fenton. On the generation of stream rating curves. *Journal of Hydrology*, 564:748–757, 2018.

[11] M. Lee et al. Construction of rating curve at high water level considering rainfall effect in a tidal river. *Journal of Hydrology: Regional Studies*, 37:100907, 2021.

[12] S. Pati et al. A reverse-stage routing scheme for estimating stage and discharge in data-scarce upstream river sections. *Journal of Hydrology*, 620:129556, 2023.

[13] S. Y. Chang, Z. Ghahremani, L. Manuel, S. M. H. Erfani, C. Shen, S. Cohen, K. J. Van Meter, J. L. Pierce, E. A. Meselhe, and E. Goharian. The geometry of flow: Advancing predictions of river geometry with multi-model machine learning. *Water Resources Research*, 59(12):e2023WR036733, 2023.

[14] Anupal Baruah, Reihaneh Zarrabi, Sagy Cohen, J. Michael Johnson, and Riley McDermott. Interpretable machine learning for predicting rating curve parameters using channel geometry and hydrological attributes across the united states. *Scientific Reports*, 15:1234, 2025.

[15] Asgeir Petersen-Øverleir and Trond Reitan. Accounting for rating curve imprecision in flood frequency analysis using likelihood-based methods. *Journal of Hydrology*, 366(1-4):89–100, 2009.

[16] J. Le Coz, B. Renard, L. Bonnifait, F. Branger, and R. Le Boursicaud. Combining hydraulic knowledge and uncertain gaugings in the estimation of hydrometric rating curves: A Bayesian approach. *Journal of Hydrology*, 509:573–587, 2014.

[17] V. Mansanarez, J. Le Coz, B. Renard, M. Lang, G. Pierrefeu, and P. Vauchel. Bayesian analysis of stage-fall-discharge rating curves and their uncertainties. *Water Resources Research*, 52(9):7424–7443, 2016.

[18] V. Mansanarez, I. K. Westerberg, N. Lam, and S. W. Lyon. Rapid stage-discharge rating curve assessment using hydraulic modeling in an uncertainty framework. *Water Resources Research*, 55(11):9765–9787, 2019.

[19] David Ocio, Nataliya Le Vine, Ida Westerberg, Florian Pappenberger, and Wouter Buytaert. The role of rating curve uncertainty in real-time flood forecasting. *Water Resources Research*, 53(5):4197–4213, 2017.

[20] I. K. Westerberg and H. K. McMillan. Uncertainty in hydrological signatures. *Hydrology and Earth System Sciences*, 19(9):3951–3968, 2015.

[21] T. Reitan and A. Petersen-Øverleir. Bayesian power-law regression with a location parameter, with applications for construction of discharge rating curves. *Stochastic Environmental Research and Risk Assessment*, 22:351–365, 2008.

[22] H. K. McMillan and I. K. Westerberg. Rating curve estimation under epistemic uncertainty. *Hydrological Processes*, 29:1873–1882, 2015.

[23] Seyed Mohammad Hassan Erfani, Mahdi Erfani, Sagy Cohen, Austin R. J. Downey, and Erfan Goharian. A large dataset of fluvial hydraulic and geometry attributes derived from USGS field measurement records. *Environmental Modelling & Software*, 180:106136, 2024.

[24] Nataliya Bulygina, Neil McIntyre, and Howard Wheater. Bayesian conditioning of a rainfall-runoff model for predicting flows in ungauged catchments and under land use changes. *Water Resources Research*, 47(2):W02503, 2011.

[25] Nataliya Bulygina and Hoshin V. Gupta. How bayesian data assimilation can be used to estimate the mathematical structure of a model. *Stochastic Environmental Research and Risk Assessment*, 24:925–937, 2010.

[26] Harry Joe. *Dependence Modeling with Copulas*. Monographs on Statistics and Applied Probability. CRC Press, Boca Raton, FL, 2014.

[27] M. Sklar. Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, 8:229–231, 1959.

[28] Mojtaba Sadegh, Hamed Moftakhari, Hoshin V. Gupta, Elisa Ragno, Omid Mazdiyasni, Brett Sanders, Richard Matthew, and Amir AghaKouchak. Multihazard scenarios for analysis of compound extreme events. *Geophysical Research Letters*, 45(11):5470–5480, 2018.

[29] Shahrbanou Madadgar and Hamid Moradkhani. A bayesian framework for probabilistic seasonal drought forecasting. *Journal of Hydrometeorology*, 14(6):1685–1705, 2013.

[30] Omid Mazdiyasni, Amir AghaKouchak, Steven J. Davis, Shahrbanou Madadgar, Ali Mehran, Elisa Ragno, Mojtaba Sadegh, Ashmita Sengupta, Subimal Ghosh, and Mohsen Niknejad. Increasing probability of mortality during Indian heat waves. *Science Advances*, 3(6):e1700066, 2017.

[31] F. Aristizabal, F. Salas, G. Petrochenkov, T. Grout, B. Avant, B. Bates, R. Spies, N. Chadwick, Z. Wills, and J. Judge. Extending Height Above Nearest Drainage to model multiple fluvial sources in flood inundation mapping applications for the U.S. National Water Model. *Water Resources Research*, 59, 2023.

[32] Christopher KI Williams and Carl Edward Rasmussen. *Gaussian processes for machine learning*, volume 2. MIT press Cambridge, MA, 2006.

[33] A. E. Sikorska and B. Renard. Calibrating a hydrological model in stage space to account for rating curve uncertainties: general framework and key challenges. *Advances in Water Resources*, 105:51–66, 2017.

[34] P. F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39:841–862, 1998.

[35] F. Laio and S. Tamea. Verification tools for probabilistic forecasts of continuous hydrological variables. *Hydrology and Earth System Sciences*, 11:1267–1277, 2007.

[36] Reihaneh Zarrabi, Riley McDermott, Seyed Mohammad Hassan Erfani, and Sagy Cohen. Bankfull and mean-flow channel geometry estimation through machine learning algorithms across the CONtiguous United States (CONUS). *Water Resources Research*, 61(2):e2024WR037997, 2025.

A Assessing Stage Variability Across USGS Gauges

To quantify how much stage varies within a narrow range of discharge, a site-level spread metric was developed using the IFMHA dataset. This metric is a descriptor of gauge behavior across CONUS and is separate from the main copula framework. It is used to diagnose whether this behavior is limited to a few sites or is noticeable at the national scale.

Measurements at each gauge were restricted to physically plausible values ($Q > 0$ and $0 < H < 20$ m) and clipped to the 1st–99th percentile range of both variables. Gauges retaining fewer than 50 observations after filtering were excluded. Discharge at each remaining gauge was divided into ten within-site deciles, and deciles containing fewer than four observations were dropped. Within each decile, the interquartile range of stage was computed, that is, the range between the 25th and 75th percentile of stage values in that decile. The site-level metric is the mean of these ten values:

$$\overline{\mathrm{IQR}}_H = \frac{1}{K} \sum_{k=1}^{K} (H_{75,k} - H_{25,k}) \tag{6}$$

where $H_{25,k}$ and $H_{75,k}$ are the 25th and 75th percentiles of stage within discharge decile k, and K is the number of retained deciles. The metric, expressed in meters, quantifies the spread of observed stage values within narrow discharge bands at each gauge. Because each band still includes a small range of discharge values, part of the observed spread may reflect this residual discharge variation rather than stage variability alone. The metric should therefore be interpreted as an approximation and is used only to place the study sites within the broader distribution of gauge behavior across the conterminous United States.

B Stochastic Rating Curve Development

B.1 Marginal distribution selection

For each variable, four parametric marginal families including Gamma, Gaussian, Truncated Gaussian, and Student-t are evaluated. The parameters are fitted using Maximum Likelihood Estimation (MLE), and the optimal family is selected via the Bayesian Information Criterion (BIC) as follows:

$$\mathrm{BIC} = k \ln(n) - 2 \ln(\hat{L}), \tag{7}$$

where k is the number of parameters, n the sample size, and $\hat{L}$ the maximized likelihood. The model with the lowest BIC is selected, balancing goodness of fit against model complexity.

B.2 Copula selection and conditional distribution

Once the marginal models are identified, observations are mapped to the unit interval through the probability integral transform (PIT), $u = F_Q(Q)$ and $v = F_H(H)$. Five bivariate copula families are then considered for the dependence structure between Q and H: Gaussian, Student-t, Clayton, Frank, and Gumbel, with parameters estimated via MLE and family selection by BIC. Following Equation 5, $f(H \mid Q)$ is obtained by combining the fitted copula density $c(u, v)$ with the marginal stage density $f_H(H)$. The probabilistic rating curve is then defined by selected percentiles of the resulting distribution.

B.3 Performance evaluation

BIC, above, selects the best-fitting model among candidates; it is not a measure of predictive performance. Three separate metrics are used to evaluate predictive performance: NSE, empirical coverage, and spread. The Nash–Sutcliffe Efficiency (NSE) is a deterministic performance metric. It evaluates the accuracy of the median (P50) stage prediction against observations, without accounting for the probabilistic spread of the forecast:

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{n}(H_{\mathrm{obs},i} - \hat{H}_{p50,i})^2}{\sum_{i=1}^{n}(H_{\mathrm{obs},i} - \bar{H}_{\mathrm{obs}})^2}, \tag{8}$$

where $H_{\mathrm{obs},i}$ is the observed stage, $\hat{H}_{p50,i}$ is the median prediction, and $\bar{H}_{\mathrm{obs}}$ is the mean observed stage. The second metric is the empirical 95% coverage, which measures the proportion of observations falling within the 95% predictive interval [34, 35].

$$\mathrm{Coverage}_{95} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\left[\hat{H}_{2.5,i} \le H_{\mathrm{obs},i} \le \hat{H}_{97.5,i}\right] \tag{9}$$

where $\hat{H}_{2.5,i}$ and $\hat{H}_{97.5,i}$ are the 2.5th and 97.5th percentile predictions, respectively. A well-calibrated model should yield a coverage close to 0.95. Coverage alone can be misleading: a very wide interval reaches 95% coverage trivially, without being informative. The third metric, spread, measures the average width of the 95% predictive interval:

$$\mathrm{Spread}_{95} = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{H}_{97.5,i} - \hat{H}_{2.5,i}\right) \tag{10}$$

A smaller spread indicates a sharper, more informative interval, provided coverage remains close to 0.95. Together, NSE, coverage, and spread provide: point-prediction accuracy (NSE), probabilistic reliability (coverage), and interval sharpness (spread).

C Case-Study Sites

C.1 Site Characteristics

Three sites from the IFMHA dataset were selected for detailed analysis: the Little Sioux River at Correctionville (USGS 06606600), the Des Moines River near Saylorville (USGS 05481650), and the Brazos River at Waco (USGS 08096500). These sites span a range of drainage areas, stream orders, and hydrologic settings, and each exhibits the stage variability described in the previous section. Figure 8 summarizes their catchment attributes alongside the discharge and stage distributions observed at each site.

At each site, only the high-flow pairs at or above bankfull discharge ($Q \ge Q_{\mathrm{bf}}$) were kept. The bankfull values come from machine-learning estimates across NHDPlus V2.1 reaches [36]. A site was retained only if it contained at least 100 pairs above bankfull, ensuring sufficient data for statistically reliable estimates. Figure 9 shows the rating curve fitted at each site alongside quantile-quantile (Q-Q) plots evaluating the fit of the marginal distributions used in the copula analysis.

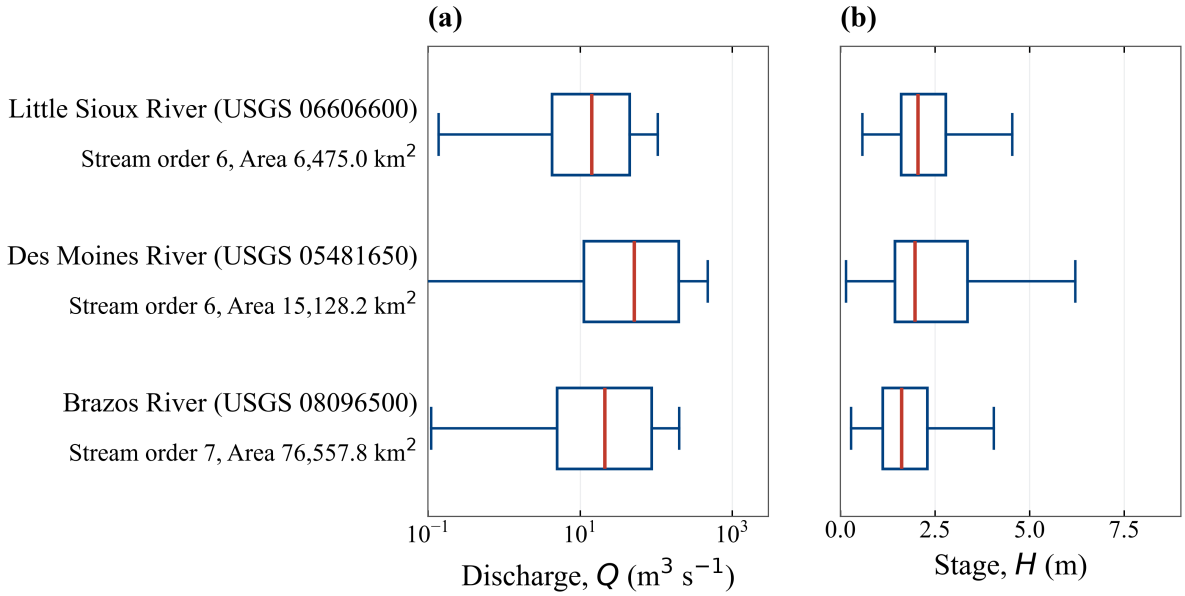


Figure 8: Discharge (a) and stage (b) distributions at the three case-study sites.

C.2 Marginal Distribution and Copula Selection

Four marginal distributions including Gamma, Gaussian, truncated Gaussian (t-Gauss), and Student's t were evaluated for each dataset. These distributions were chosen to capture the skewed, bounded, and heavy-tailed characteristics commonly observed in discharge and stage data. Discharge was strongly right-skewed at all three sites and was best characterized by the Gamma distribution in every case. The best-fit distribution for stage varied by site: Gamma provided the best fit at the two Iowa sites, whereas a truncated Gaussian best modeled the Brazos River, consistent with the bounded stage response observed during high flows (see Table 3).

Table 3: Marginal distributions, copula families, and performance metrics for each study site.

USGS Site	Marginals		Copula Model		Performance Statistics			
Number	Q	H	Family	Param (θ)	BIC	NSE	95% Cov.	95% Spr.
06606600	Gamma	Gamma	Gumbel	3.275	-278.81	0.812	95.48%	2.20 m
05481650	Gamma	Gamma	Gumbel	3.410	-323.87	0.800	94.10%	1.84 m
08096500	Gamma	t-Gauss	Frank	18.065	-468.25	0.928	97.80%	2.17 m

To model the dependence between Q and H, five copula families were evaluated: the Gaussian and Student's t from the elliptical class, and the Clayton, Frank, and Gumbel from the Archimedean class. Archimedean parameters were estimated via maximum likelihood estimation (MLE); the Gaussian correlation parameter ρ was estimated by inverting Kendall's τ using $\rho = \sin(\pi\tau/2)$. The two Iowa sites were best characterized by the Gumbel copula, whose strong upper-tail dependence reflects the tight coupling of high discharge and high stage. The Brazos River was best fit by the Frank copula, which has no tail dependence.

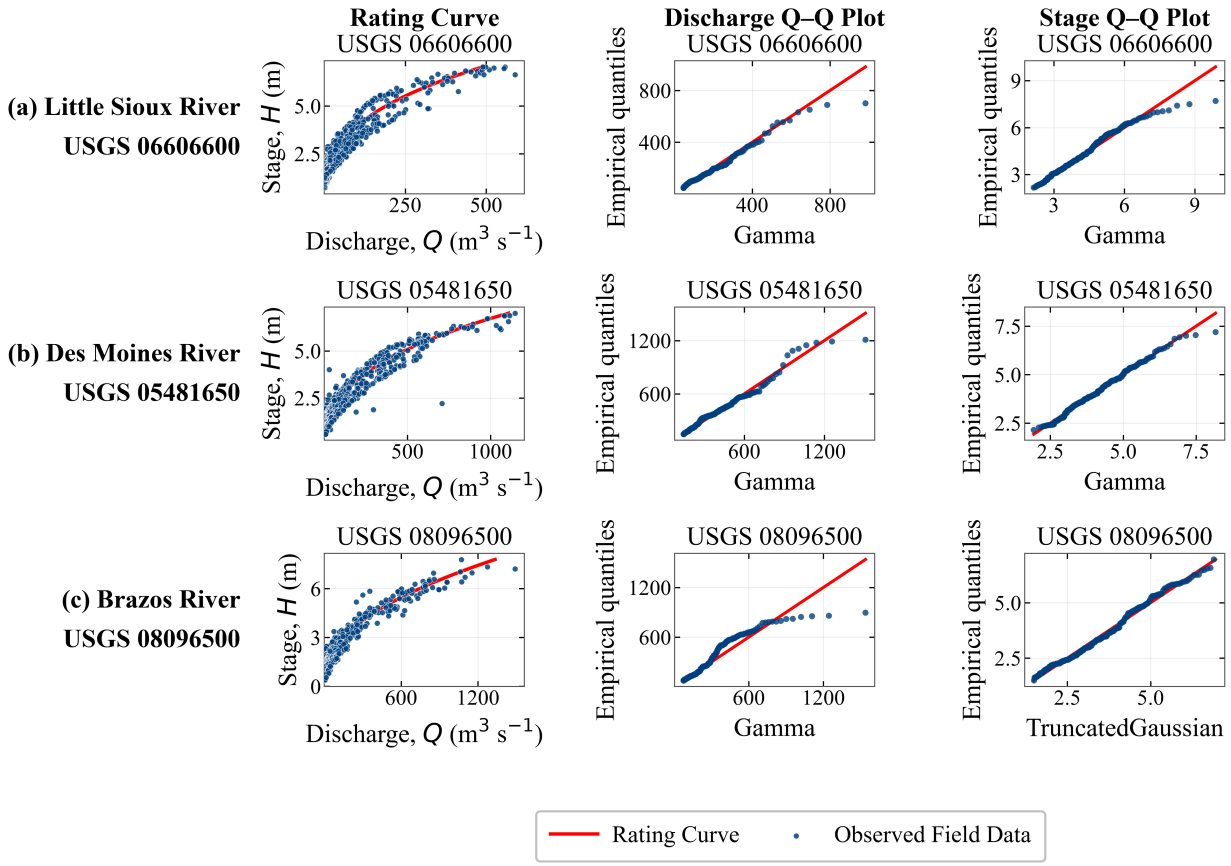


Figure 9: rating curves (left column) and Q–Q plots of the fitted marginal distributions for discharge (center) and stage (right).